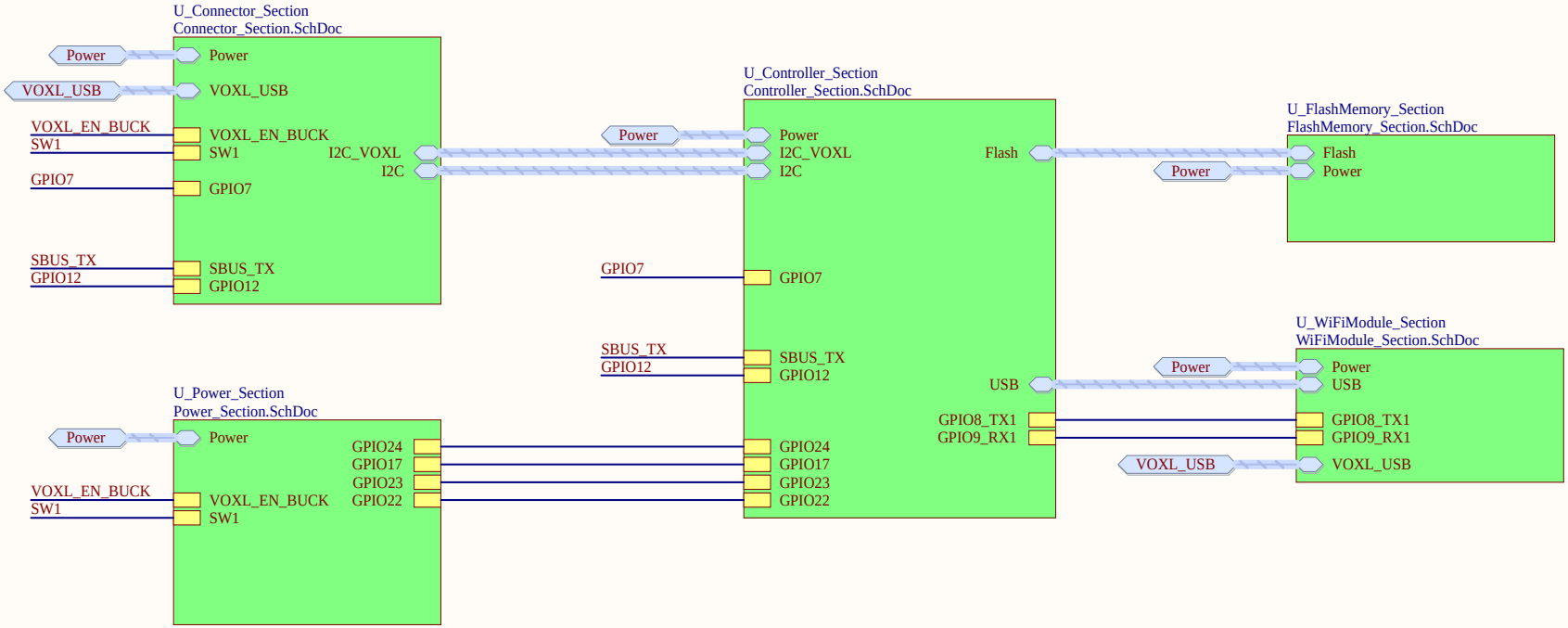
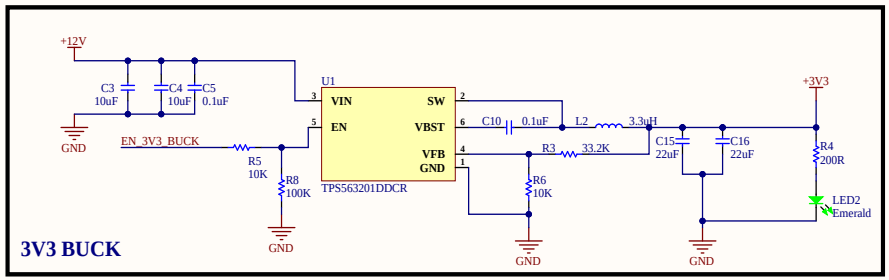
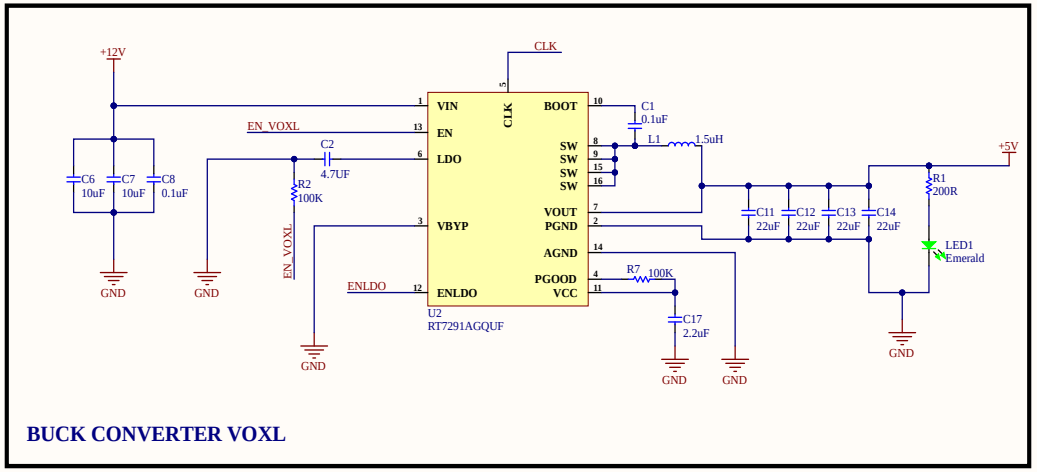
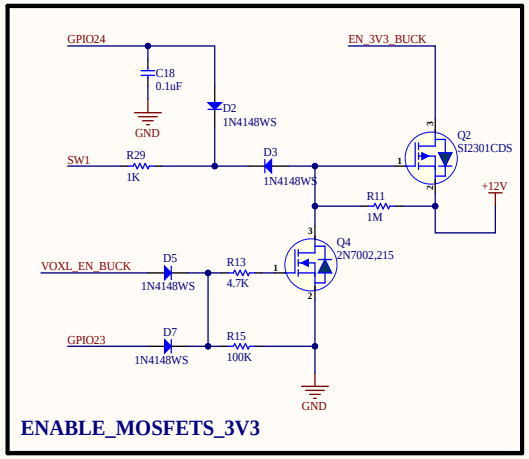


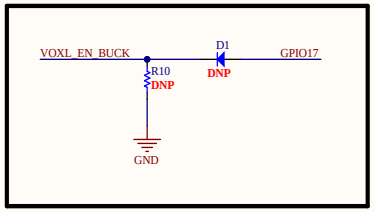

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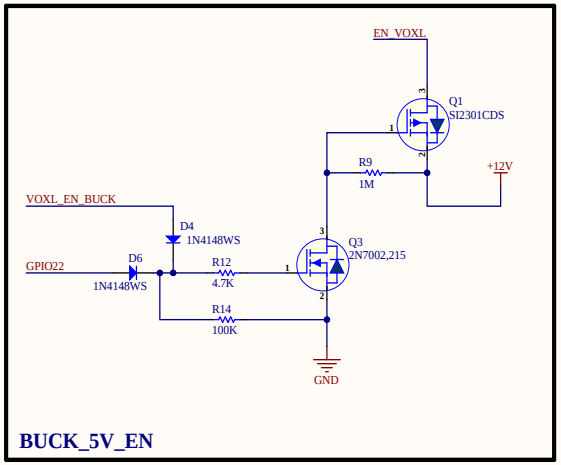
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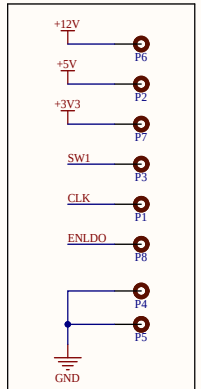


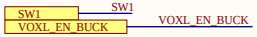



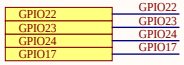







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| File: Powermodule-v7.0.2.PrjPcb | |
| Date: 02/05/2024 | Page Num: 2 |
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## TESTPOINTS

GPIO6 — P9
GPIO1 — P10
GPIO0 — P11
GPIO2 — P12
GPIO3 — P13
GPIO4 — P14
GPIO5 — P15
GPIO26_ADC0 — P16
GPIO27_ADC1 — P17
GPIO28_ADC2 — P18
GPIO29_ADC3 — P19
GPIO10 — P34
GPIO11 — P35
GPIO16 — P36
GPIO18 — P37
GPIO19 — P38
GPIO25 — P39

RUN — P26 (RUN)

SWCLK — P24
SWDIO — P25
DEBUG

U3 — RP2040TR7

| | | | |
|---|---|---|---|
| 1 | IOVDD_1 | GND | 57 |
| 2 | GPIO0 | QSPI_SS_N | 56 QSPI_SS |
| 3 | GPIO1 | QSPI_SD1 | 55 QSPI_SD1 |
| 4 | GPIO2 | QSPI_SD2 | 54 QSPI_SD2 |
| 5 | GPIO3 | QSPI_SD0 | 53 QSPI_SD0 |
| 6 | GPIO4 | QSPI_SCLK | 52 QSPI_SLCK |
| 7 | GPIO5 | QSPI_SD3 | 51 QSPI_SD3 |
| 8 | GPIO6 | DVDD_2 | 50 |
| 9 | GPIO7 | IOVDD_6 | 49 |
| 10 | IOVDD_2 | USB_VDD | 48 |
| 11 | GPIO8 | USB_DP | 47 USB_D_P |
| 12 | GPIO9 | USB_DM | 46 USB_D_N |
| 13 | GPIO10 | VREG_VOUT | 45 |
| 14 | GPIO11 | VREG_VIN | 44 |
| 15 | GPIO12 | ADC_AVDD | 43 ADC_AVDD |
| 16 | GPIO13 | IOVDD_5 | 42 |
| 17 | GPIO14 | GPIO29_/_ADC3 | 41 GPIO29_ADC3 |
| 18 | GPIO15 | GPIO28_/_ADC2 | 40 GPIO28_ADC2 |
| 19 | TESTEN | GPIO27_/_ADC1 | 39 GPIO27_ADC1 |
| 20 | XIN | GPIO26_/_ADC0 | 38 GPIO26_ADC0 |
| 21 | XOUT | GPIO25 | 37 GPIO25 |
| 22 | IOVDD_3 | GPIO24 | 36 GPIO24 |
| 23 | DVDD_1 | GPIO23 | 35 GPIO23 |
| 24 | SWCLK | GPIO22 | 34 GPIO22 |
| 25 | SWDIO | IOVDD_4 | 33 |
| 26 | RUN | GPIO21 | 32 SCL0 |
| 27 | GPIO16 | GPIO20 | 31 SDA0 |
| 28 | GPIO17 | GPIO19 | 30 GPIO19 |
| | | GPIO18 | 29 GPIO18 |

+3V3
C19 0.1uF
C21 0.1uF
C24 0.1uF
+1V1
C26 0.1uF
C20 0.1uF
C22 0.1uF
C23 0.1uF
C25 0.1uF
+1V1
GND

GPIO17 WAS MISPRINTED AS GPIO28
Checked it with previous design and Datasheet

### HEARBEAT_LED

GPIO25 — R16 200R — LED3 Emerald — GND

### USB_CON

USB_D_P, USB_D_N
R18 27R — D_N
R20 27R — D_P

### ADC_REF

+3V3 — R17 200R — ADC_AVDD — C31 2.2uF — GND

### INTERNAL_VREG_STABL.

+1V1, +3V3
C29 2.2uF
C30 2.2uF
GND

### CRYSTAL

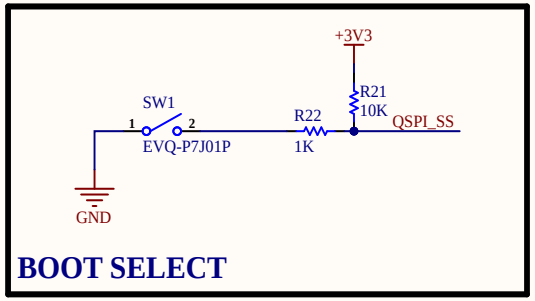
C27 10pF — X1 (12MHz) — C28 10pF
R19 1K
XOUT
XIN
GND

SBUS_TX — GPIO12
GPIO22, GPIO23, GPIO24, GPIO17
GPIO12, GPIO7
GPIO8_TX1, GPIO9_RX1

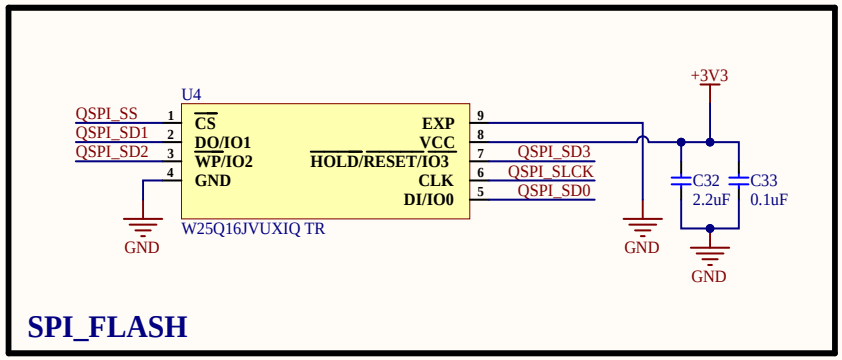
I2C: SDA0, SCL0
I2C_VOXL: SDA_VOXL, SCL_VOXL
USB: D_N, D_P
Power: +12V, +5V, +3V3, GND
Flash: QSPI_SLCK, QSPI_SS, QSPI_SD1, QSPI_SD0, QSPI_SD3, QSPI_SD2

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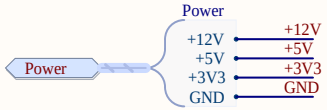
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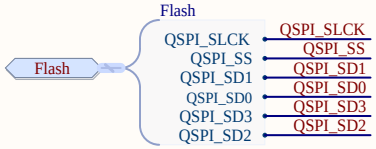



**BOOT SELECT**



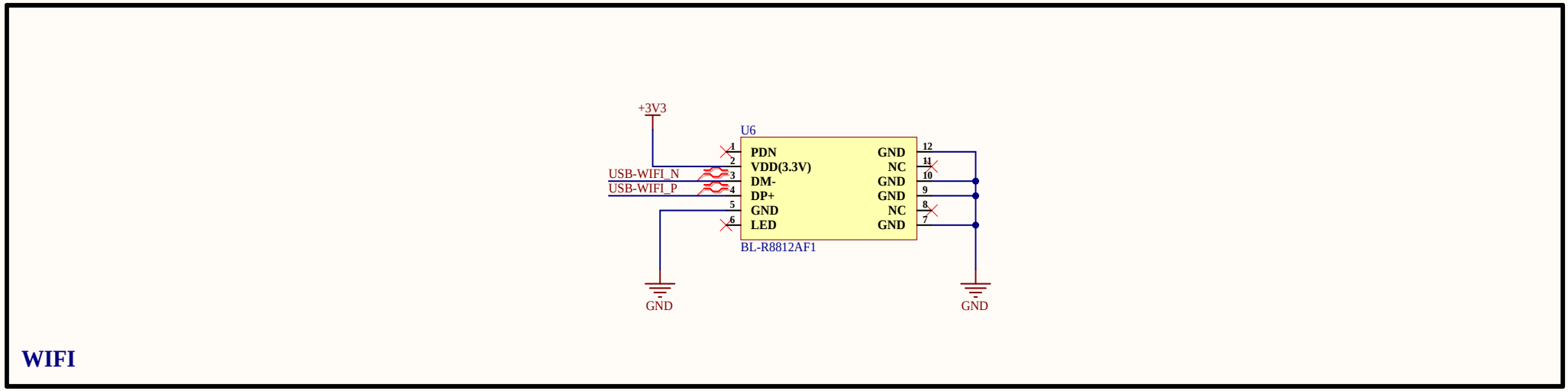

**SPI_FLASH**

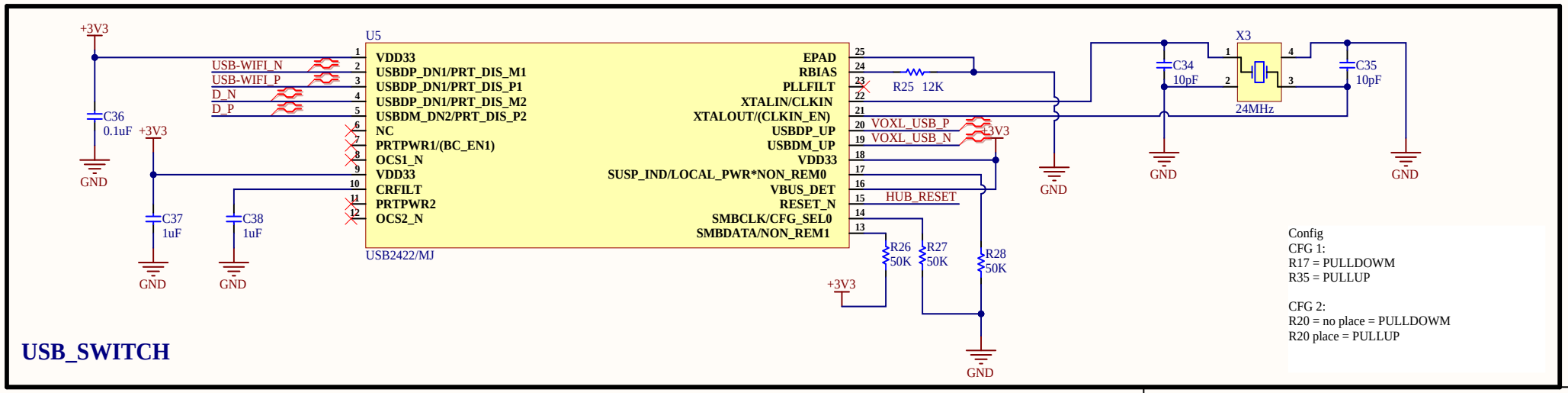


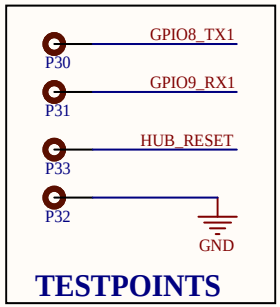




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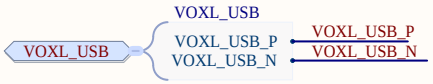


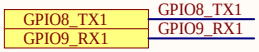



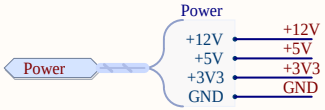







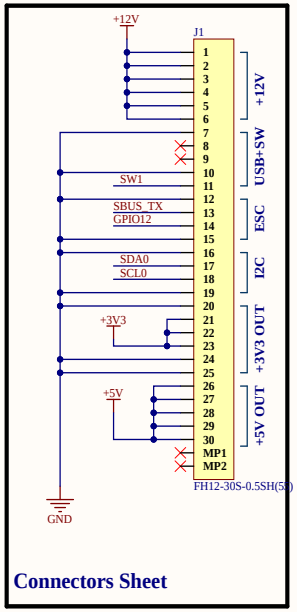


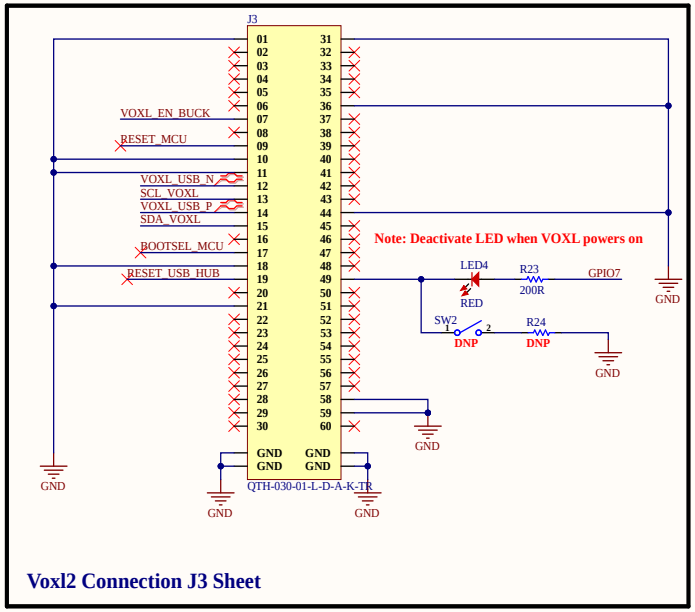




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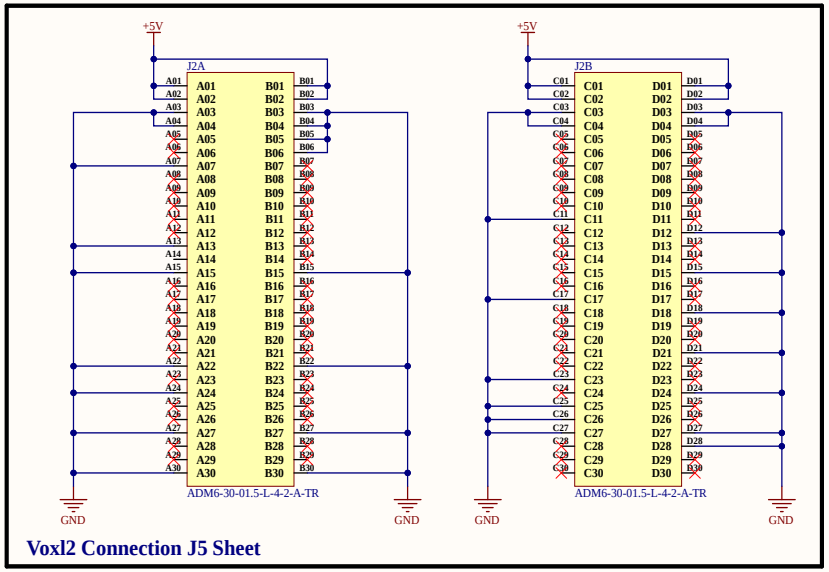
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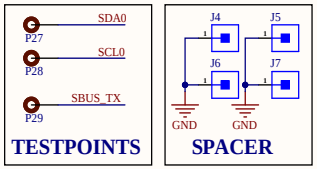

Connectors Sheet




Voxl2 Connection J3 Sheet



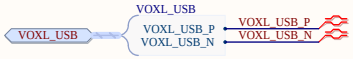

Voxl2 Connection J5 Sheet

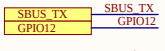


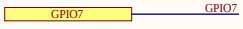

TESTPOINTS




SPACER



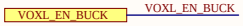













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